

Sociolinguistic Properties of Word Embeddings

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Introduction

Just a single word can prime our thinking and initiate a stream of associations. Not only do words denote concepts or things-in-the-world; they can also be combined to convey complex ideas and evoke connotations and even feelings. The meanings of our words are also contextual — they vary across time, space, linguistic communities, interactions, and even specific sentences.

As rich as words are to us, in their raw form they are merely a sequence of phonemes or characters. Computers typically encounter words in their written form. How can this sequence become meaningful to computers? Word embeddings are one approach to represent word meanings numerically. This representation enables computers to process language semantically as well as syntactically. Embeddings gained popularity because the “meaning” they captured corresponds to human meanings in unprecedented ways. In this chapter, we illustrate these surprising correspondences at length.

Because word embeddings capture meaning so effectively, they are a key ingredient in a variety of downstream tasks involving natural language (Artetxe et al., 2017; Young et al., 2018). They are used ubiquitously in tasks like translating languages (Artetxe et al., 2017; Mikolov, Tomas, Quoc V. Le, and Ilya Sutskever, 2013) and parsing clinical notes (Ching et al., 2018).

But embeddings are more than just tools for computers to represent words. Word embeddings can represent *human* meanings because they learn the meanings of words from *human* language use — such as news, books, crawling the web, or even television and movie scripts. They thus provide an invaluable tool for learning about ourselves, and exploring sociolinguistic aspects of language at scale. For example, embeddings have shown how our language follows statistical laws as it changes across time (Hamilton et al., 2016), and how different languages share similar meanings (Mikolov, Tomas, Quoc V. Le, and Ilya Sutskever, 2013). The similarities between the representation of semantic information in humans and embeddings have even provided theoretical insight into how we learn, store and process meaning (Günther et al., 2019).

At the same time, any pejorative meaning in human language, like gender stereotypes, are also learned by these models. What’s worse, these meanings may be amplified in downstream applications of word embeddings (Bolukbasi, T., Chang, K. W., Zou, J. Y., Saligrama, V., & Kalai, A. T., 2016; Dixon et al., 2018; Ethayarajh et al., 2019). This has now forced us to confront such biases head on, fueling conversations about human biases and stigma while also providing new opportunities for intervention (Bolukbasi, T., Chang, K. W., Zou, J. Y., Saligrama, V., & Kalai, A. T., 2016; Dixon et al., 2018; Manzini et al., 2019; Zhao et al., 2017).

After a brief introduction to embeddings, we describe empirical evidence showing how meanings captured in word embeddings correspond to human meanings. We then review the theoretical evidence illustrating how word embeddings correspond to — and diverge from — human cognitive strategies to represent and process meaning. In turn, these divergences illustrate challenges and future research directions in embeddings.

Part 1. Brief introduction to word embeddings

The term “word embedding” denotes a set of approaches to quantify and represent the meanings of words. These approaches represent the meaning of a word w in the vocabulary V as a vector (an array of numbers). Not all vector representations are embeddings, however. In the simplest vector representation, each word w in the vocabulary maps onto a distinct vector v_w which has a “1” in a single position (marking out that word) and zeroes everywhere else (Table 1). This is called “one-hot” encoding and relies on an essentially symbolic (and arbitrary) correspondence between the word and the vector.

Table 1. Example One-Hot Encoded Vectors					
Vocabulary Word	A	And	Animal	...	Zoo
One-hot vector for the word “And”	0	0	1	...	0
One-hot vector for the word “Zoo”	0	0	0	...	1

Word embeddings, by contrast, represent words as vectors derived from co-occurrence patterns. In the simplest word embedding, each word w in a corpus is represented as a V -dimensional vector, and the element V_j in the representation of w corresponds to the number of times w co-occurs with the j -th vocabulary word in the corpus. Vocabulary sizes can range from tens of thousands to hundreds of thousands of words, depending on the corpus, and many words do not co-occur. Thus, vectors end up long and sparse. While these vectors encode some semantic information, they do so inefficiently; they are just as long as one-hot encoding, they require more memory to specify, and they fail to exploit the latent meanings shared between words, which could be used to compress the representation.

Contemporary word embeddings exploit the strategy of *distributed* representation ([Rumelhart et al. 1986](#)). In a distributed representation, a concept (like “woman”) is represented by a unique pattern of connection to a limited number of low-level units (e.g., neurons, or artificial neurons). These low-level units are shared across concepts; by the power of combinatorics, a relatively small number of low-level units can encode an enormous number of distinct concepts. Words still correspond to vectors, but the components of the vector now represent the “weight” of connection to the corresponding low-level unit. This encoding is much more efficient; each word is now represented using a much smaller number of dimensions (e.g., 100-500) rather than the

full V dimensions. Because they are now standard, we only discuss distributed embeddings in this chapter, and refer to them as “embeddings” for brevity.

There are two core approaches to arrive at embeddings (Baroni et al., 2014). The first is count-based. Such approaches begin with the co-occurrence matrix computed from the corpus and attempt to reduce its dimensionality by finding latent, lower-dimensional features that encode most of its structure. Latent Semantic Analysis (LSA), for example, performs dimensionality reduction (Singular Value Decomposition) on a term-document matrix,¹ to yield lower-dimensional vector representations of words and documents (Landauer and Dumais 2008). A more recent count-based method today is GloVe (“**G**lobal **V**ectors”). Given two words w_i and w_j , GloVe tries to find embeddings that minimize the difference between the product of the corresponding word embeddings and the log probability of their co-occurrence in the corpus, using weighted least squares (Pennington et al., 2014).

A second core approach uses an artificial neural network (ANN) architecture to learn word embeddings from a given corpus as the by-product of some prediction task. For example, word2vec learns word-vectors either by predicting some set of context words given some target word (SkipGram) or by predicting a target word given its surrounding context words (CBOW) (Mikolov et al., 2013; Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J, 2013). The vector for word w corresponds to the weights from a one-hot encoded representation of w to the hidden layer of a shallow ANN; these weights (and hence vectors) are updated as the ANN is trained on the CBOW or Skip Gram prediction task. Vectors may be initially random, but across many iterations of the prediction task, training the network gradually improves how well the word-vectors capture word meanings.

Both approaches to learn embeddings relies on the Distributional² Hypothesis (Firth, 1957; Harris, 1981). This hypothesis suggests that “you shall know a word by the company it keeps,” or, that words with similar meanings tend to occur in similar contexts (Firth, 1957). Context may be defined as a sentence, a fixed window of words, or even a document. In predictive models for word embeddings (e.g., word2vec) word-vectors are learned by predicting words from their context (or vice versa), while in count-based models for word embeddings (e.g., GloVe) word-vectors are derived from co-occurrences between words. Even if two words do not directly co-occur in any contexts, they may have similar representations because they share higher order contexts. For example, even if “policewoman” and “police officer,” never co-occur in a corpus, they will have similar representations insofar as they are both predicted by similar words, such as “crime” and “law.” In practice, certain count based and prediction models can produce vectors with similar performance on downstream tasks if the hyperparameters are correctly tuned

¹ This is a matrix containing word counts per document (or other context). The matrix contains a row for each vocabulary word and a column for each document.

² Describing a model as distributional refers to the distributional patterns of words and the use of the Distributional Hypothesis mentioned earlier, *not* to whether words are modeled as distributed representations. Contemporary word embeddings are distributional but also use distributed representations.

(Sanjeev Arora et al., 2016; Levy & Goldberg, 2014). This is remarkable considering that count based models begin with global information about word co-occurrences, while predictive models implicitly learn this information from local co-occurrences.

Variants of word2vec, such as FastText ([Bojanowski et al. 2017](#)) and BERT ([Devlin, J., Chang, M. W., Lee, K., & ...](#)), learn vectors not only for words³ but also for sub-word information. They might learn, for example, the meaning of the suffix “-ing,” which is used in English to form gerunds or present participles, as in running, swimming, and walking. This additional flexibility allows these approaches to represent words outside of the initial corpus vocabulary, since a new word often contains similar morphemes to words already inside the model vocabulary. Embedding methods like BERT (“**B**idirectional **E**ncoder **R**epresentations from **T**ransformers”) depart from previous methods in that they produce *contextualized* embeddings. That is, in trained BERT models, each word in each sentence is mapped to its own distinct embedding. Representations for words are contextualized in that the vector for a word in a given instance is slightly adjusted by information from nearby words. BERT also combines multiple hidden layers in an ANN (e.g., by averaging or concatenating them) to produce a final representation of a word or sentence. This approach is motivated by the fact that different kinds of meaning are represented at different layers in an ANN.

Various other more specialized frameworks for embeddings exist as well, which are tailored to specific outcomes or linguistic properties. For example, models may incorporate additional information in the prediction task beyond the text to yield embeddings that are specialized for a downstream task, such as predicting sentiment so that words are well suited for sentiment analysis (Tang et al., 2014). Further, hyperbolic word embeddings have been created to enable word embeddings to encode hypernymy (Nickel & Kiela, 2017). Given the variety of embedding frameworks, this chapter focuses on the core models and their properties.

Embeddings reach their top performance when they are given constraints on the size of the space in which they map meaning (Gladkova et al., 2016; Mikolov et al., 2013; Yin et al., 2018). In count-based word embeddings, reducing the matrix of co-occurrences forces words to be represented over latent semantic dimensions. Predictive models are similarly only given a limited number of dimensions (again, often a few hundred in word2vec and FastText) in which to map words, forcing them to discover latent semantic dimensions rather than simply memorize each mapping. This compression is thought to be critical to the success of word embeddings (Sanjeev Arora et al., 2016).

³ We use the term “words” in this chapter for readability, but embeddings may use any token (unit of meaning) not only words. For example, as part of data pre-processing, researchers often transform commonly occurring phrases into single tokens (e.g., “New_York”), so that vectors are learned for these phrases as well.

Part 2. Are word embeddings good semantic representations? The empirical evidence

2.1 Similarity in word embeddings and human-rated similarity

Embedding models trained with the approaches described in the previous section provide uncanny correspondences to human-rated semantics in empirical studies. Most crucially, words which are rated as more semantically similar by humans also tend to have similar word-vectors. Similarities between word-vectors in a trained model may be measured with cosine similarity, which ranges between -1 and 1. Cosine similarity measures how similar, or, equivalently, *close* in space two words are. Cosine similarities between pairs of word-vectors in embeddings often carry strong correlations with how humans rate the similarity of these same pairs of words. For example, a model trained SkipGram reaches correlations between .7 and .8 with tuned hyperparameters correlate with how similar humans rate these same pairs of words (Finkelstein et al., 2002; Levy et al., 2015). Word embeddings also perform well on semantic tasks that indirectly involve semantic similarity, such as categorizing nouns, and syntactic and semantic analogy tasks (Mikolov et al., 2013; Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J, 2013). Indeed, word2vec gained fame for its ability to solve the analogy “woman:man::king:?”

Of course, cosine-similarity between two word-vectors is only an estimate of semantic similarity. In particular, cosine similarity blurs semantic similarity and semantic *relatedness*. Cosine similarity does not distinguish when two words capture similar concepts from broader the notion of semantic relatedness. For example, while “truck” and “van” are similar concepts, “van” and “driver” are not similar concepts, but they are strongly associated and related to one another (Lenci, 2018). Further, cosine similarity weights all dimensions equally – thus, for example, while a human might rate pairs of antonyms as *dissimilar*, antonyms often have high cosine similarity in semantic space. Indeed, two words are antonyms because they are very similar in all aspects of their meaning except for one salient aspect. In practice, this means that they are used in very similar contexts (Lenci, 2018). For example, the opposite pair “woman” and “man” are similar in many regards (e.g., both describe people and are singular rather than plural), so a word embedding learns these as very similar. However, these two words are opposites in gender. As humans, we weigh the gender aspect of their meaning as very *salient*, overlooking all their similarities and thus considering them *dissimilar*. In fact, the notion of semantic similarity itself is vague. There are many *types* of semantic relations between words that can drive up the similarity between two words, such as hypernyms, antonyms, and synonyms.

One way to overcome this vague interpretation of cosine similarity between two words is to compare their similarity along specific semantic *features*, also called dimensions (Grand, G., Blank, I. A., Pereira, F., & Fedorenko, E, 2018). For example, a researcher might compare two words' meanings in terms of gender, wealth, sentiment, or size. Approaches to do this, described next, offer very powerful tools working with embeddings, and also shows additional ways in which embeddings capture human meaning.

2.2 Semantic features in word embeddings

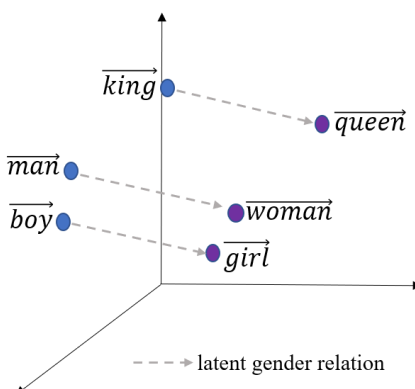
There are two broad approaches often used to look at semantic features using word embeddings. Both are deductive and require the selection of “anchor” words that represent the feature. The first approach is to simply compute cosine similarity between an anchor word-vector representing that feature (e.g., “woman”) and some outcome words (e.g., career words) which we’re investigating for gendered meanings. For example, to compare how two occupational titles (e.g., “scientist” and “nurse”) are gendered, we can compare their cosine similarities to “woman.” We can build on this by comparing outcome words’ cosine similarities to *two* anchor words, rather than one, each of which represents the two poles of a feature (e.g., “woman” and “man” to represent the feature for gender) (Caliskan et al. 2017; Jones et al. 2020; Lewis and Lupyan 2020). In particular, Caliskan et. al. compared the cosine similarity between “man” and stereotypically masculine occupations with the cosine similarity between “woman” and stereotypically feminine occupations (Caliskan et al., 2017), and thus quantified how word embeddings learned these words as stereotypical gendered.

The second approach to examine semantic features applies specifically to features like gender, which may be represented as a bipolar relation (e.g., femininity v.s. masculinity). We describe this approach next.

2.2.1 Relations as lines in semantic space

A striking empirical finding in embeddings is that they encode relations, like *gender*, in the learned semantic space. While words are represented as points in semantic space, relational concepts, like gender, are *lines* in this space (Bolukbasi, T., Chang, K. W., Zou, J. Y., Saligrama, V., & Kalai, A. T., 2016; Grand, G., Blank, I. A., Pereira, F., & Fedorenko, E, 2018; Kozlowski et al., 2019). These lines point to the poles of the relation they represent; for example, a line corresponding to gender points to femininity in one direction, and to masculinity in the other direction. Put another way, this dimension corresponds to the displacement, or shift in meaning, from one point (femininity) to another point (masculinity). Remarkably, this shift in meaning is similar for the movement from “queen” to “king,” and “girl” to “boy,” visualized in Figure 1. Of course, embedding models are not told any meanings of gender or other semantic features explicitly — these are *learned* from the patterned ways in which words appear in contexts.

Figure 1. Visualizing a Latent Gender Relation



A straightforward way to extract a relation in a trained model is to simply subtract the anchor words from each other. For example, we can subtract “man” from “woman” to get a line corresponding to gender. Conceptually, if we assume that the latent meanings of woman and man are largely equivalent (both human, adults, nouns, singular, etc.) except for their opposite gender components, then subtraction cancels out all but the gender differences across each component. Since word-vectors are noisy, we can use a few examples of word-vectors for each pole to get a better estimate (e.g., average out “woman,” “girl,” “her,” and “she” to get a better prototype of a feminine word-vector). Using a variant of this method, Bolukbasi et al (Bolukbasi, T., Chang, K. W., Zou, J. Y., Saligrama, V., & Kalai, A. T., 2016) performed dimensionality reduction (Principal Components Analysis) on a matrix composed of feminine word-vectors and masculine word-vectors. They then found that the first principal component explained most of the variation in these vectors and corresponded to a “gender” line.

After extracting this line, we can use it to compute how words carry meaning with respect to this semantic feature. For example, we can take a new word, like “boyish,” and the cosine similarity between boyish and this gender dimension tells us how “boyish” is gendered. This is a scalar ranging from -1 to 1. The sign of this scalar tells us whether the word lies on the masculine or feminine end of the dimension, and the magnitude tells us how strongly the word lies on feminine or masculine end (i.e., how strongly the word connotes men or women). As described next, these processes to extract semantic features can be used to extract a variety of meanings, such as morality, social class, and pleasantness, and meanings extracted with these methods correspond to human ratings of these meanings ([Bolukbasi, T., Chang, K. W., Zou, J. ...; Grand, G., Blank, I. A., Pereira, F.,...; Kozłowski et al. 2019](#)).

2.2.2 Semantic features correspond to human meaning and large-scale cultural patterns

Using the above approaches, various studies illustrate how words correspond to humans’ meanings along specific semantic dimensions and large-scale cultural patterns. These studies use large, commonly available pre-trained embeddings or their training corpora, such as Google News, web data (Common Crawl), and Google Books. First, using a pre-trained model on GoogleNews, Bolukbasi and colleagues computed cosine similarities of stereotypically gendered words with a line corresponding to gender, where crowd-workers rated stereotypicality. They find that these words, like “nurse” and “doctor,” were gendered in stereotypical ways in the embedding (Bolukbasi, T., Chang, K. W., Zou, J. Y., Saligrama, V., & Kalai, A. T., 2016). They replicated results using a pretrained model on Common Crawl data. Caliskan and colleagues examined meanings surrounding gender, race and sentiment, comparing findings with those from findings from implicit association tests. They, too, show that results hold in both pre-trained models of GoogleNews and Common Crawl. They find, for example, that stereotypical European names are more pleasant than stereotypical African American names, and that math is more masculine than feminine, while art is more feminine than masculine (Caliskan et al., 2017). Finally, Kozłowski and colleagues examine dimensions such as gender, race and class in the Common Crawl model and other pretrained models, comparing results to survey responses from crowd workers. They find that the ways in which words carry meanings of gender, race, and

class in these embeddings strongly correlate how humans rate meanings of these words with respect to gender, race, and class (Kozlowski et al., 2019).

Meaning isn't just encoded in language but is also tightly intertwined with real-world patterns and changes. Embeddings reflect this relationship. For example, the gendered meanings of names in embeddings (using pretrained models on Web Crawl and Google News), tend to match up with Census data on the gender of individuals with those names, with a correlation of .84 (Caliskan et al., 2017). Further, the gender and class meanings of occupational titles (using embeddings trained on Google NGrams data) change across time in ways that correspond to historical trends, such as timelines for industrialization (Kozlowski et al., 2019). For example, Kozlowski and colleagues find that "cotton" connotes affluence in embeddings trained on Google Books from 1810-1910 from the U.S., matching the fact that the cotton industry underlied the economy of the U.S. South for much of these years. Strikingly, this connotation does not hold among embeddings trained on books from Great Britain. Finally, a variety of studies have shown that the way occupations are gendered in embedding models (trained on a variety of data) corresponds very tightly (correlations around .9) to the proportion of women in these occupations (Bolukbasi, T., Chang, K. W., Zou, J. Y., Saligrama, V., & Kalai, A. T., 2016). In particular, Garg and colleagues showed how occupations' gendering in embeddings trained on Google Books from 1900-2007 even change across time in ways consistent with Census data on the percentages of women and men in these occupations (Garg et al., 2018). Alongside other recent work with embeddings ([e.g., Joseph and Morgan 2020; Lewis and Lupyan 2020; Jones et al. 2020](#)), the above scholarship suggests that the meanings encoded by embeddings carry considerable external validity. These correspondences also underscore that "meaning" in language is connected with both meanings widely shared across individuals, and material, on-the-ground cultural patterns.

Thus far, only *correlations* between language and individuals' meaning, and between language and large-scale patterns have been shown using embeddings. However, the applications of embeddings described throughout the chapter paves the way for causal investigation into the relationship between language, thought, culture and material patterns ([Lewis and Lupyan 2020](#)). For example, future research might investigate the *lag* structure between language and cultural changes. The nature of these relationships between language, thought, and culture are core questions in social science.

Certain patterns of meaning found in word embeddings — such as that embedding models contain stereotypically gendered occupations — are pejorative. When these pejorative meanings involve protected attributes like gender and race, they are often described as machine-learned bias (Bolukbasi, T., Chang, K. W., Zou, J. Y., Saligrama, V., & Kalai, A. T., 2016; Caliskan et al., 2017; Manzini et al., 2019; Zhao et al., 2019). Many tasks utilizing embeddings use existing, commonly used pretrained embedding models, or the same corpora on which these models are trained (e.g, Google news, Wikipedia, Twitter). Thus, studies and applications using these corpora and models can propagate biases. For certain words, embeddings may even learn exaggerated biases compared to what is in the raw text (Ethayarajh et al., 2019). Of course, this bias begins with humans; embeddings simply learn from data we produce. A variety of methods are now being developed to eliminate these biases from trained models, offering opportunities for *intervention* into these biases (Zhang et al., 2018; Zhao et al., 2018). Still, what differentiates "bias" from "meaning," is subjective and contested. Thus, like other machine-learning tools,

embeddings have stirred important conversations about *what* is bias, *where* it is and how to confront it in *both* humans and machines.

2.3 Word embeddings and neural data

Thus far, all correspondences between human and embeddings described in this chapter have come in the form of conscious ratings from humans. However, embeddings also correspond to our unconsciously activated meanings upon reading and hearing language; in the form of neural activations. Embeddings are certainly not the first semantic representation (derived from text) which correspond to neural data (e.g., (Mitchell et al., 2008)). However, embeddings have offered new empirical insights on how semantic information is stored and processed in our own brains and have been shown to correspond beyond just the level of words. For example, Vodrahalli and colleagues map fMRI responses to watching movie scenes, and representations of these scenes derived from word embeddings (Vodrahalli et al., 2018). Specifically, to arrive at these representations they obtain human-annotations summarizing each scene, and then take the weighted sum of word embeddings in the annotation (S. Arora et al., 2017), where word-embeddings are pre-trained on Wikipedia. Further, given fMRI data of participants reading stories, Dehghani and colleagues could identify *which* story participants are reading from embedding representations of the stories (Dehghani et al., 2017). In fact, they found that this was possible across languages — whether the original language of the story (and embedding training data) was English, Mandarin, or Farsi. This suggests these embedding methods captured the “gist” or meaning of the story in a way that transcended language and specific words. Such correspondences between embeddings and neural data not only highlight that embeddings are useful tools for neuro-semantic research but also provide additional validity to embeddings.

Part 3. What kind of “meaning” do word embeddings really capture?

Like most computational methods to represent the meaning of words, word embeddings rely on the Distributional Hypothesis (Firth, 1957; Harris, 1981), as described earlier. Thus, meanings encoded in word-vectors are derived from *context* — or, distributional patterns in language. While studies using embeddings often cite this hypothesis, they implicitly interpret “meaning” in this hypothesis in various ways, falling into two camps (Lenci 2008). First, that the distributional patterns of a word *correlate* to the (latent) meaning of a word, “whatever” meaning may be, and thus can tell us *about* the (latent) meaning of a word. Second, in a stronger reading, that the meaning of a word *is* (at least in part) its distributional patterns. This stronger reading is cognitive and causal, suggesting a mechanism (language) by which humans themselves learn the meaning of words. Of course, even the stronger reading does not mean that meaning is *only* learned relationally and from language, just that we *can* learn meaning in this way.

The meaning captured in embeddings, and derived from distributional patterns, should be distinguished from other ways in which words can be meaningful. While embeddings, like humans, learn statistical information about words, humans can additionally have a *grounded* understanding of certain words (Bryson, 2008). For example, humans, like embeddings, can very accurately know when and how to use the word “dog,” “bark,” and “bump.” But humans can

also use these words to identify real referents of dogs, barks, and bumps. And humans may have an embodied understanding of words — that is, the physical senses words correspond to like the physical feeling of “bump” and the sound of “bark.” In fact, some work has tried to begin building in some degree of “grounding” to models, through multimodal embeddings (Bruni et al., 2014; Feng, 2010). These integrate information derived from text with features extracted from images or other modalities. Of course, not all words can be grounded in physical realities, and for many words, even humans only know how and when to use words. For instance, while concrete concepts are more likely learned from sensorimotor experience, abstract concepts like “freedom” cannot be learned through such experience. Thus, distributional models and a stronger reading of the Distributional Hypothesis suggests a *mechanism* (language) for learning these more abstract concepts (Borghi et al. 2017; Günther et al. 2019).

3.1 The “meaning” of semantic features

Using these methods to look for relations in semantic space, like gender, involves several implicit assumptions about meaning. First, measuring meaning requires the selection of anchor words to represent either end of the dimension. To select anchor words, researchers either choose representative words (Kozłowski et al., 2019), use crowd-workers (Bolukbasi, T., Chang, K. W., Zou, J. Y., Saligrama, V., & Kalai, A. T., 2016) and/or use established lexicon to select representative words (Jones et al., 2020; Sagi & Dehghani, 2014). Selecting anchor word-vectors to represent a concept, like “femininity” may be subjective, results may vary depending on what anchor words are chosen and how they are used (Ethayarajh et al., 2019).

Second, these methods may operationalize meaning as taking a particular *form*. Measuring gender as a line, for example, assumes that meanings of gender range continuously from one pole to the other (e.g., masculinity to femininity) and assumes that having more feminine meaning corresponds to less masculine meanings, and vice versa. While comparing cosine similarities between anchor words does not assume that more femininity implies less masculinity, this approach still does look for gendered meanings as oriented by two poles. This particular form of meaning is called a binary opposition (Levi-Strauss, 2008).

While a binary opposition is certainly a fundamental structure of meaning, it comes in various subtypes. And, it is certainly not the only possible form for meaning. Meaning may be hierarchical, continuous, dichotomous, gradable (Geeraerts, 2009, p. 87). For example, temperature may be hot/cold, or hot/temperate/cold, or hot/warm/temperate/cool/cold. Other oppositions might be discrete and mutually exclusive, such that one pole necessitates the lack of the other (e.g., arrive/leave, or alive/dead). Meaning may even be pieces to larger structures of meaning composed of oppositions. For example, Western meaning-system for direction consists of two binary oppositions of north/south and east/west (or, of three oppositions: left/right, up/down, and front/back) (Geeraerts, 2009, p. 87).

Unraveling these assumptions behind these methods to look for relations also offers insight into how to measure meaning structures beyond word (points) and relations (lines). For example, race is not easily reduced to a binary opposition. Meanings of race may be better characterized as a system of binary racial oppositions such as Black/white and Black/Asian. Further, some meanings derived from embeddings have higher correspondences to human-ratings than do other meanings (Grand, G., Blank, I. A., Pereira, F., & Fedorenko, E, 2018) — namely, gender (Joseph

& Morgan, 2020). Perhaps certain meanings better fit the assumptions imposed by these methods (Ethayarajh et al., 2019). Future research might examine the types of meanings and relations that are better measured in embeddings, and compare methods to measure them.

Importantly, the fact that meaning takes these forms in our semantic system does not imply that their corresponding constructs in the real world take this same form. For example, even if gendered may be measured as a binary opposition in semantic space in a way that explains the variation of “gendered” meanings in a corpus, this does not imply anything about the distribution of popular conceptions of gender identities or about gender identity itself.

The fact that word embeddings encode relational features, like gender, also reflects that embeddings encode some degree of compositionality. Compositionality is the notion that some meaning can be broken into pieces or recombined, and is a crucial aspect of human meaning. Relations like gender illustrate compositionality of meaning because they are identified and manipulated using vector arithmetic: subtracting and adding word-vectors, as described earlier.

In fact, embeddings are commonly evaluated on how *well* they learn relational features like gender. This comes in the form of an analogy test. For example, the Google Analogy Test which includes a series of syntactic and semantic analogies divided up into subsections, including world capitals, countries’ currencies, family, tense and plurality (Mikolov et al., 2013; Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J, 2013). For example, an analogy in the semantic section of World Capitals will include: “Berlin: Germany as Paris: _____,” and if the model responds correctly it will return: “France.” In the syntax section with tense, a sample analogy might be “Walk: Walked as Run: Ran.” Thus, common benchmarks for measuring the quality of trained embeddings (analogy tests) *require* that the model has learned some degree of compositionality, in the form of latent relations between entities such as capitals and tense.

Beyond relational features like gender, embeddings exhibit compositionality in other ways as well. For example, sentences are well represented by simply a weighted sum of their constituent word embeddings (S. Arora et al., 2017). Further, models described earlier that use sub-word information, like FastText, also enable compositionality in that new words can be represented. For example, if the original FastText model has learned “Twitter” and “-ify,” it can form a representation for “twitterify.” Given the importance of compositionality to human meaning, the fact that embeddings encode some degree of compositionality highlights a consistency between meaning in humans and embeddings.

3.2 Cognitive underpinnings of word embeddings

The fact that human meaning correlates to meaning captured in word embeddings suggests that word embeddings may be a good empirical *description* of our own semantic representations, even with the limitations described earlier in this chapter. Further, the fact that word embeddings represent words as *distributed* representations, where each word corresponds to the activation of lower level units (e.g. artificial neurons) suggests that embeddings provide a description of meaning that is also theoretically consistent with the way humans represent meaning (Günther et al., 2019).

Predictive word embeddings additionally provide certain theoretical consistencies with how humans *learn* and *process* semantic representations (Günther et al., 2019). For example, predictive word embeddings learn word-vectors through attempts to predict missing target words in a given context. Similarly, prediction is a powerful learning mechanism for humans and we learn by predicting missing information in our environment (Lupyan & Clark, 2015; Rumelhart, 2017), including words given contexts (DeLong et al., 2005). While the similarities between embeddings and human cognition should not be overstated, these similarities can provide insight into how language is processed by humans, and could be processed by machines.

Embeddings were not developed according to any model of online (i.e., real-time) processing of words. However, recently Arora et al. proposed that word embeddings follow a generative model, much like how topic models assume a generative model for topics (Sanjeev Arora et al., 2016, 2018). In Arora et al.'s framework, words in a corpus are generated by a slow, random walk in semantic space. At any time point in the corpus t , a “discourse vector” represents what is being talked about. Like a topic in a topic model, each discourse vector c defines a probability distribution over words. That is, it defines how likely each word in the vocabulary is to be emitted given the current context. This is generative in that it models how words in a corpus are produced in real-time.

The cognitive underpinnings of Arora's generative model are similar to those of topic models, and even suggest how embeddings might be understood as topic models. To process a sentence, we must retrieve many concepts from memory given a continuous input of information. To facilitate this retrieval, we use the current *gist* of what we are reading or seeing and predict most *relevant* concepts (Griffiths et al., 2007). Arora's model operationalizes the “gist” as the discourse vector at some time point. Given a window of words, the “gist” that generated these words may be estimated as the average of constituent word-vectors.⁴ Arora further suggests that there are important, recurring “gists:” these building blocks of meaning are analogous to topics from LDA topic modeling. In practice, Arora's model produces fine-grained topics; for example, Arora et al. describe the five closest words to one topic, derived from an embedding trained on Wikipedia: “orchestra,” “philharmonic,” “philharmonia,” “conductor,” and “symphony.” Another topic from the same model is “membrane,” “mitochondria,” “cytosol,” “cytoplasm,” and “membranes” (Sanjeev Arora et al., 2018). While words and features are commonly investigated in embeddings, Arora's model paves the way for research on topical structures in embeddings. It also provides new insights into how embeddings may be envisioned in a generative model with parallels to how humans process language.

3.3 How stable, vs contextual, is meaning?

Until this point, this chapter has largely described meaning as something invariant to context. While there must be some degree of stability in meanings to enable communication, meaning is also highly contextual. The meanings of words in a corpus depends on the context in which that corpus is generated — such as the time, place, and even attributes of the author. And even within a corpus, the meaning of a word is contextual in that it depends on the specific phrases or

⁴ In fact, Arora shows that this estimate is also the same as the CBOW task in word2vec with negative sampling, aside from rescaling. CBOW predicts a target word given the average of the word-vectors in a context window (i.e., predicts a target word given an estimated “gist”).

sentences in which it resides. And finally, meaning is an interactive process: meaning is not only evoked by a corpus, but also evoked in an individual writer and reader.

Meaning depends on the context in which the corpus is published. An often cited example is how the word “gay” carried very different meanings over the 20th century, moving from a term about emotion to one about sexual orientation (Hamilton et al., 2016). As word embeddings learn meaning in their training corpus, and meaning may be specific to a corpus, this has direct implications for using pre-trained embeddings. Using pre-trained embeddings assumes that the outcome of interest is not sensitive to the training corpus’ context — such as whether the corpus is composed of news or books, written in first person or third person, and written in the 1980s or 2000s. In some cases, using high-quality, pre-trained word-vectors from a published model, like GoogleNews, may be reasonable. Or better yet, it may make sense to seed a model with pre-trained word-vectors (rather than random word-vectors) and fine-tune the word-vectors on the corpus. This can save time and enable word embedding methods to be used even on small corpora sizes. However, in other cases, it is important to consider language in context in which it is generated – particularly, for example, when the goal is to learn about meaning, culture, and language itself.

Meaning also depends on who produces and interprets the corpus, and *their* attributes, such as their social class, race, gender. Indeed, this is well documented in annotation tasks in natural language processing, where researchers often find low agreement between annotators (Artstein & Poesio, 2008). Some research using embeddings has begun to investigate the role of interpretants to text. For example, Garten and colleagues investigate the interplay between demographics and language to jointly explore how the text of moral vignettes, and readers’ demographics, predict readers’ moral interpretations of the vignettes (Garten et al., 2019). The fact that meaning can vary widely between individuals also suggests that in some cases, *validation* of embeddings should account for demographics of raters, so that an appropriate, corresponding benchmark is used.

Meaning further varies within a corpus. In the extreme case, polysemous words can evoke very different meanings in different contexts: the meaning of “depression” is different in the context of mental health versus finance. But contexts also modulate words in more subtle ways. The word “depression [finance]” has further specific meanings in the context of “The Great Depression,” which refers to a specific, and especially notorious, economic depression. Similarly, the word “depression [mental health]” carries specific meaning in clinical contexts, where it implies a diagnosis made by a clinician following standardized diagnostic criteria. In fact, one critique of popular validation metrics for embeddings is that they rarely account for the fact that words can carry multiple meanings (Faruqui et al., 2016). Similarly, a limitation of models like word2vec, FastText, and GloVe is that they use a single word-vector for each vocabulary word in the corpus. These models smooth over the nuanced, but patterned ways we use words like “depression.” Contextualized embeddings, like BERT, offer solutions to maintain words’ contexts. But such models also come with far higher complexity than those with single vectors per word, as they represent each instance of a word. And, arguably, while words are certainly modulated by context, many of our words’ meanings are somewhat stable and patterned across contexts – otherwise, we couldn’t communicate.

Conclusion

As we have shown in this chapter, word embeddings provide powerful tools to quantitatively capture semantic information of words. At the same time, they have also inspired a wealth of empirical insights and theoretical questions on language, cognition, and meaning. In many ways, word embeddings reflect the way our own meanings of words and concepts. In other ways, they diverge.

This chapter has also highlighted several core directions for future research. For example, our theoretical understanding of the geometry of word embeddings is still underdeveloped, despite recent headway (Sanjeev Arora et al., 2016, 2018). A better understanding of the geometry of word embeddings could also shed light on how to disentangle the various types of semantic relations encoded by word embeddings, such as antonyms vs synonyms. Indeed, such relations are fundamental to language. Other burgeoning research areas include how to sensitize embeddings to linguistic context and account for polysemy, and how to ground word embeddings, such that they can learn meaning not only from words, but from other mediums like images. Additionally, many questions remain on how to validate embedding models and statistically analyze the results they produce (Ethayarajh et al., 2019; Faruqui et al., 2016; Schnabel et al., 2015). While the affinities between how embeddings and humans *process* words as distributed representations are briefly discussed in this chapter, this area also remains nascent and ripe for further investigation. Finally, an emerging territory of research is using embeddings as a *tool* to learn about our own culture and language. This includes, but is not limited to, mapping out our stereotypes and biases. Burgeoning text data sources provide cultural traces across time and place (Bail, 2014), and now word embeddings provide powerful new methods to quantify meaning in these cultural traces.

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